

Objectives

- Analyze and select an appropriate algorithmic framework for predicting student dropout risk at scale.
- Evaluate computational trade-offs between complex ML classifiers and a distributed rule-based heuristic model.
- Build an interpretable, highly scalable predictive baseline for the 1,000,000-row dataset.

Tasks Completed

Rule-Based Heuristic Model:

A rule-based heuristic model was developed after evaluating various predictive approaches for large-scale data processing. Weighted scores were assigned to identified risk and protective factors, and percentile-based thresholds were established to classify students into Low, Medium, and High Risk categories. The resulting classifications were exported for use in subsequent machine learning phases.

Machine Learning Pipeline:

A distributed machine learning pipeline was implemented using PySpark MLlib. Key academic and psychological indicators were combined into a unified feature vector through a VectorAssembler, enabling efficient feature preparation and model execution within the distributed environment.

Tasks to be Completed

N/A

Results and Analysis

Heuristic Model Outcomes:

The heuristic model successfully processed 1,000,000 student records using Spark's distributed architecture without memory-related issues. Analysis indicated that burnout and stress were among the strongest contributors to dropout risk. The model also effectively segmented students into risk categories, generating labeled data that served as a foundation for machine learning model development.

Machine Learning Pipeline Outcomes:

The PySpark MLlib pipeline was successfully executed on the processed dataset. Normalized academic and psychological indicators were incorporated into the model, allowing it to identify relationships between key risk factors and dropout outcomes. The successful implementation of the pipeline marked the transition from exploratory analysis to predictive modeling.

Challenges and Solutions

A key challenge encountered during this phase was preparing the preprocessed dataset for compatibility with PySpark MLlib's machine learning algorithms, which require multiple input features to be represented as a single feature vector. To address this issue, the VectorAssembler component was implemented to automatically combine the normalized academic and psychological indicators into a unified feature vector. This transformation ensured that the dataset met the input requirements of the machine learning pipeline and enabled efficient distributed model training within the Spark environment.

Conclusion

The objective of this phase was successfully achieved through the implementation and execution of a machine learning algorithm using big data techniques. Leveraging the large-scale dataset prepared during the previous phase, a distributed classification model was trained using relevant academic and psychological indicators. The successful integration of machine learning into the data pipeline enabled automated dropout risk prediction and advanced the project from data preparation and exploratory analysis toward model evaluation and eventual deployment.